

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Crop Recommendation Input	Accepts soil and environmental parameters	UI Layer	Done	
2	Data Validation	Validates user input before sending to model	Backend (Flask)	Done	
3	Crop Prediction	Predicts best suitable crop using ML model	ML Model	Done	
4	Dataset Processing	Loads and processes agricultural dataset	Data Layer	Done	
5	Model Training	Trains Logistic Regression / KNN / Random Forest models	ML Pipeline	Done	
6	Result Display	Shows predicted crop on web interface	UI Layer	Done	
7	Model Storage	Saves trained model using pickle	Deployment	Done	

Feature Summary:

Metric	Count / Value
Total Features Planned	7
Total Features Implemented	7
Core / Must-Have Features	Crop prediction, data preprocessing, ML model training, user input handling

Additional / Nice-to-Have Features	K-Means clustering, visualization, model comparison, Flask UI integration
Features Tested & Verified	

Feature Category Breakdown:

<i>S.No</i>	<i>Category</i>	<i>Features in Category</i>	<i>Example Features</i>
1	User Interface (UI)	Web-based input and output system	HTML form for NPK input, result display page
2	Backend / Logic	Flask application and request handling	Flask routes, prediction API
3	Database / Storage	Data cleaning and preprocessing	Handling missing values, scaling, encoding
4	API / Integration	Model training and prediction	Logistic Regression, KNN, Random Forest
5	Security / Authentication	Data analysis and graphs	Seaborn plots, Matplotlib charts